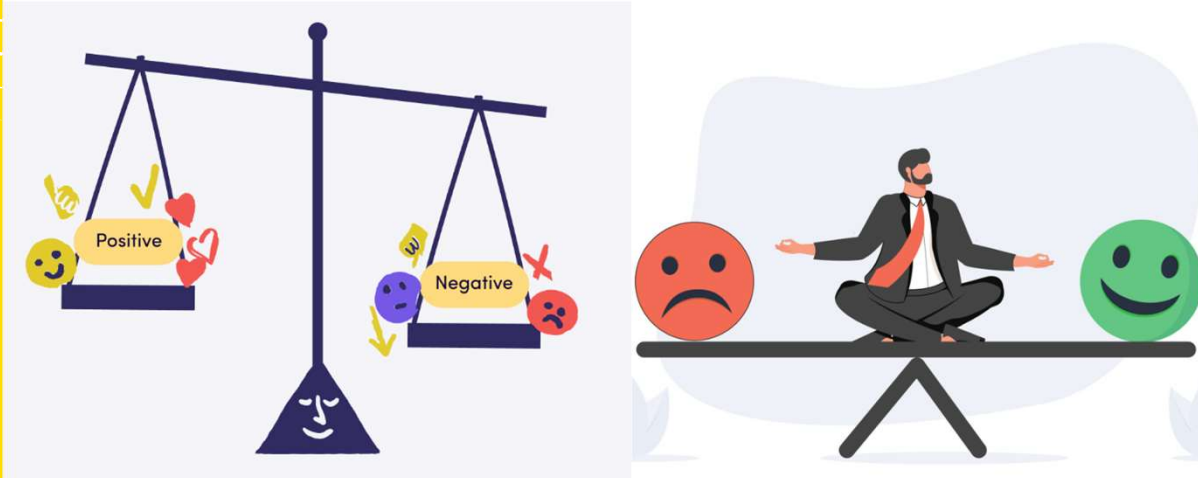


Building Grouping and Soft Actor Routing in MAPPO for CityLearn Energy Control



***Progress Report 2
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Supervisor: Flora Salim***

Fixed groups cannot follow changing states

Two problems

LEFT FROM REPORT 1

TarMAC did not balance
local and shared information well

K-means fixed each building
to one actor for the whole month



SOFT ROUTER

**Choose a complete actor from the
current building state, not only from
a fixed group**

Main innovation

The task and learning base stay the same

25 homes, one district target, grouped MAPPO

[11, 16]

25 HOMES

**Battery + HVAC
actions**

GROUPED MAPPO

**Centralized critic
Decentralized actors**

ONE TARGET

**Track district load
Keep homes
comfortable**

Main metrics: CV-RMSE, NMBE, degree-hours, comfort exceedance

$$R_{i,t} = - \left[\frac{w_{\text{nmbe}} B_t + w_{\text{cv_rmse}} E_t}{N} + w_{\text{comfort}} (w_{\text{binary}} I_{i,t} + w_{\text{degree}} d_{i,t}) \right].$$

Two upgrades build stronger actors

Useful support

[5, 18]

**TarMAC
Hybrid**

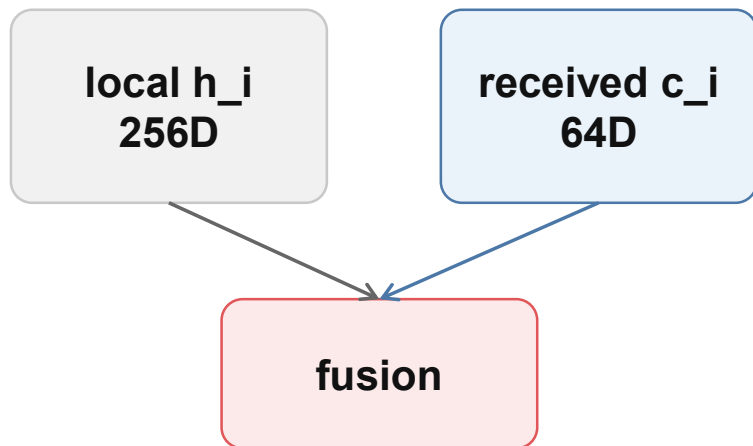
**Agglomerative
grouping**

These steps create better fixed experts for the router

TarMAC Hybrid fixes the fusion imbalance

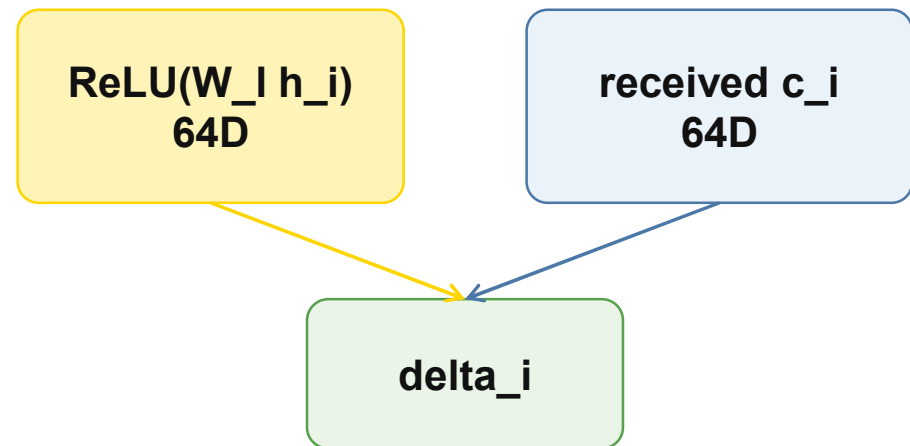
Compact local projection + nonlinear filtering + residual update

Original TarMAC



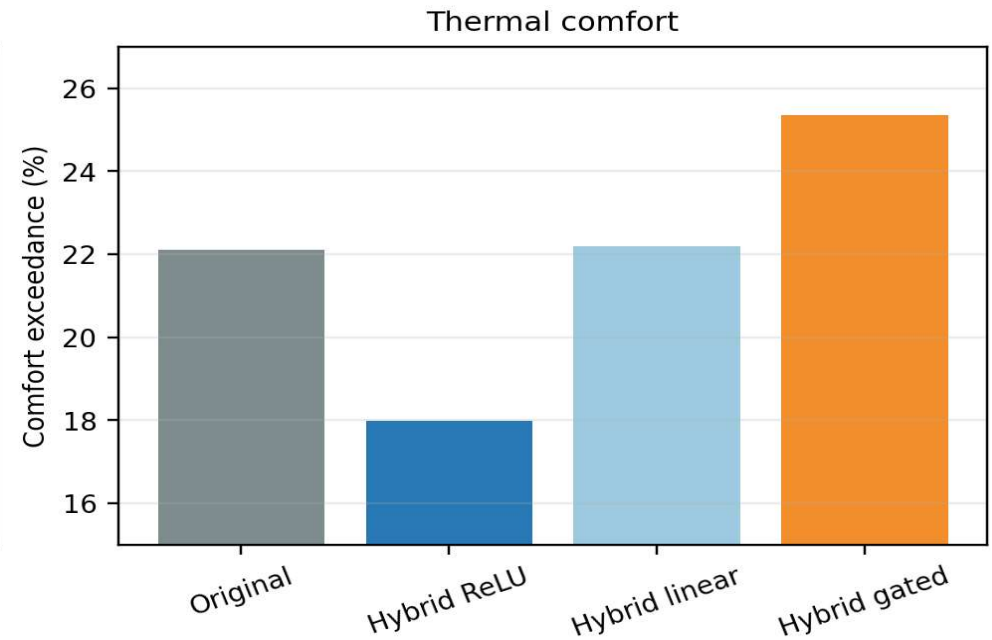
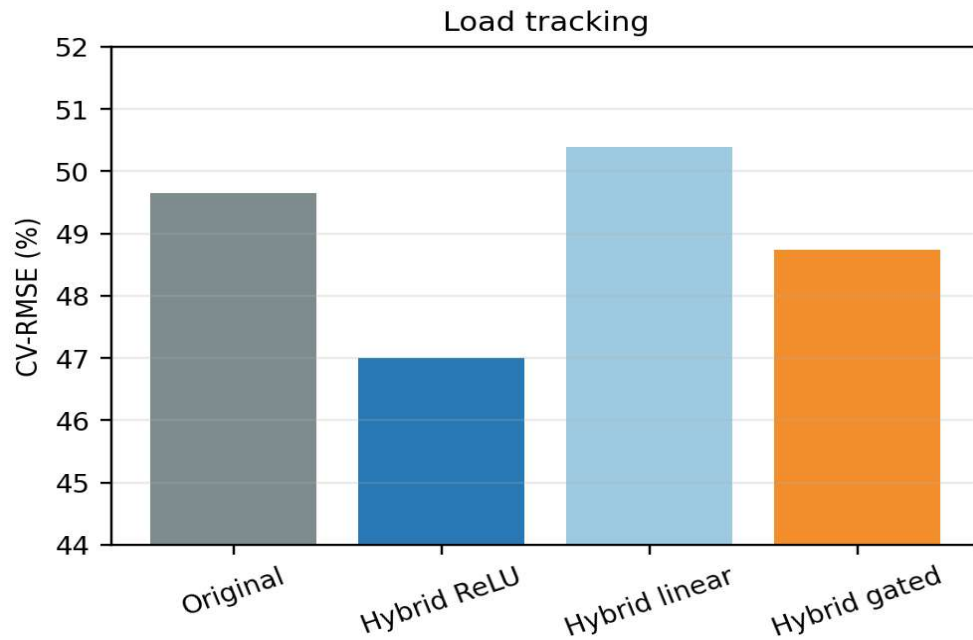
256D local path will dominate

TarMAC Hybrid



$$h'_i = h_i + \text{delta}_i$$

Hybrid improves tracking and comfort



-5.3%

CV-RMSE vs original

-18.6%

comfort exceedance

-21.4%

reward magnitude

Grouping changes what actors can learn

Four unsupervised methods

K-means

Fast centre-based
baseline
Random starts; uneven
sizes

GMM

Probabilistic
overlapping groups
Flexible but data-hungry

Agglomerative

**Deterministic Ward
merging
No random
initialization**

Balanced spectral

Similarity graph +
Hungarian assignment
Sizes differ by at most
one

Same features | same seed | same MAPPO training | same evaluation

Four Method average value across multiple seeds

Method	CV-RMSE	NMBE	DH/day	Comfort
K-means	51.56	2.53	4.57	17.33
GMM	53.08	0.94	5.90	23.75
Agglomerative	46.93	0.49	6.70	23.13
Balanced spectral	52.01	1.99	5.74	21.22

I proposed 3f from physical reasoning

The hypothesis before the main comparison

[13]

A useful building description needs three different roles

BES CAPACITY

**How much energy
can move**

HEATING MEAN

**Typical controllable
thermal need**

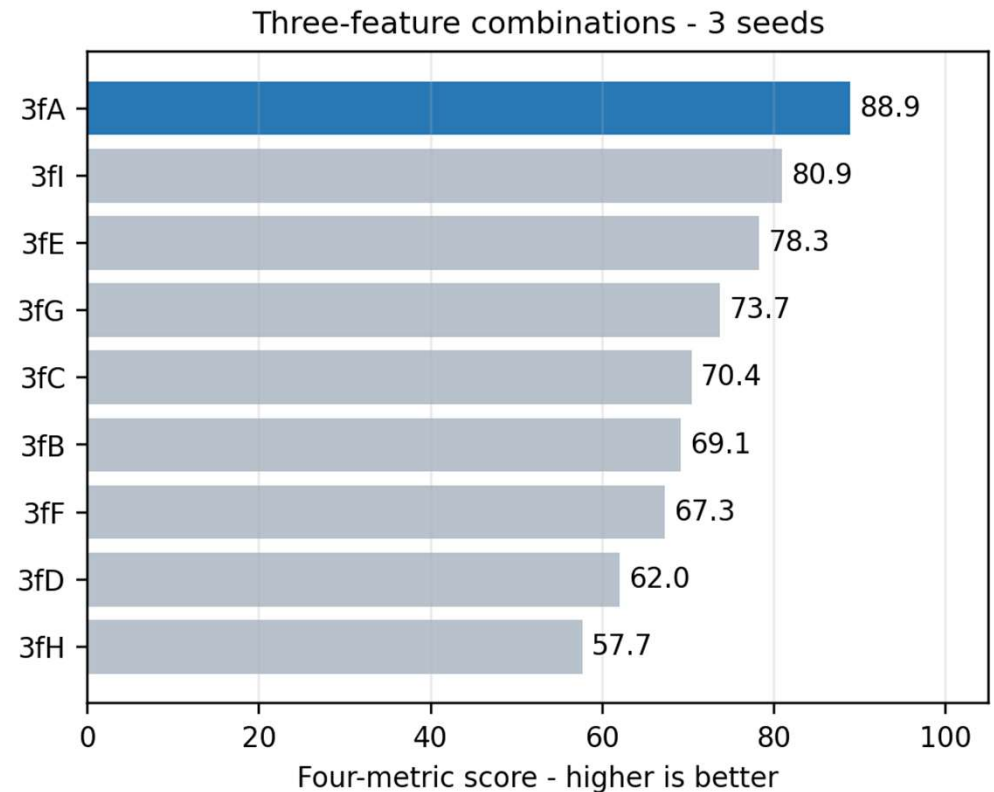
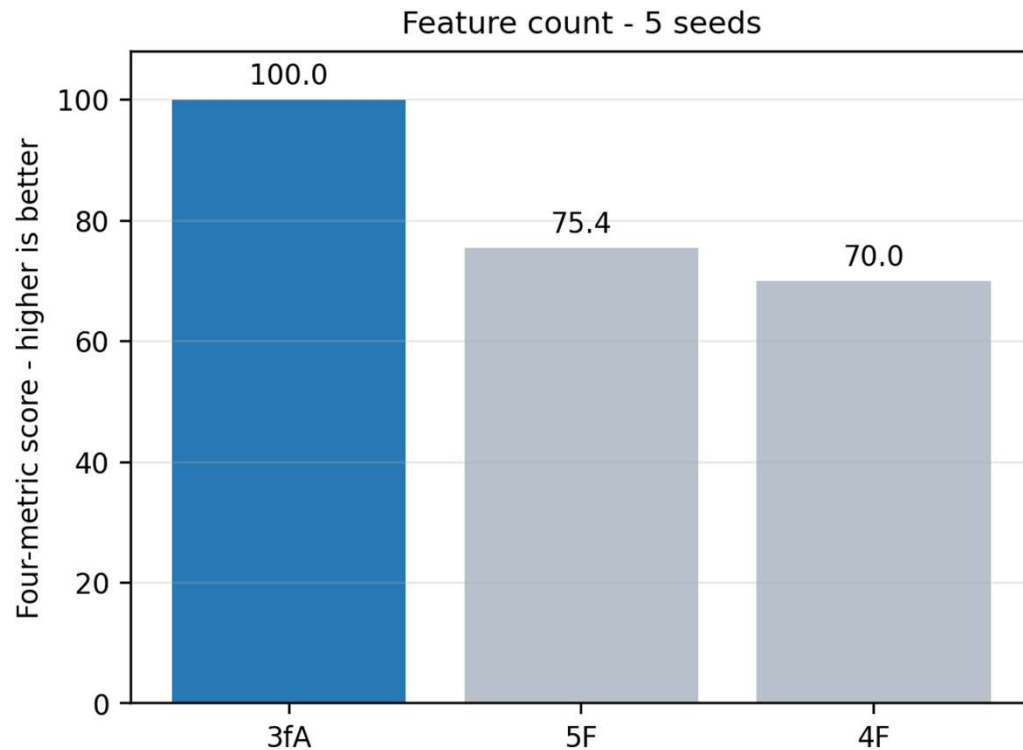
NSL MEAN

**Typical load that
cannot move**

**3fA hypothesis: clear roles + low repeated
information**

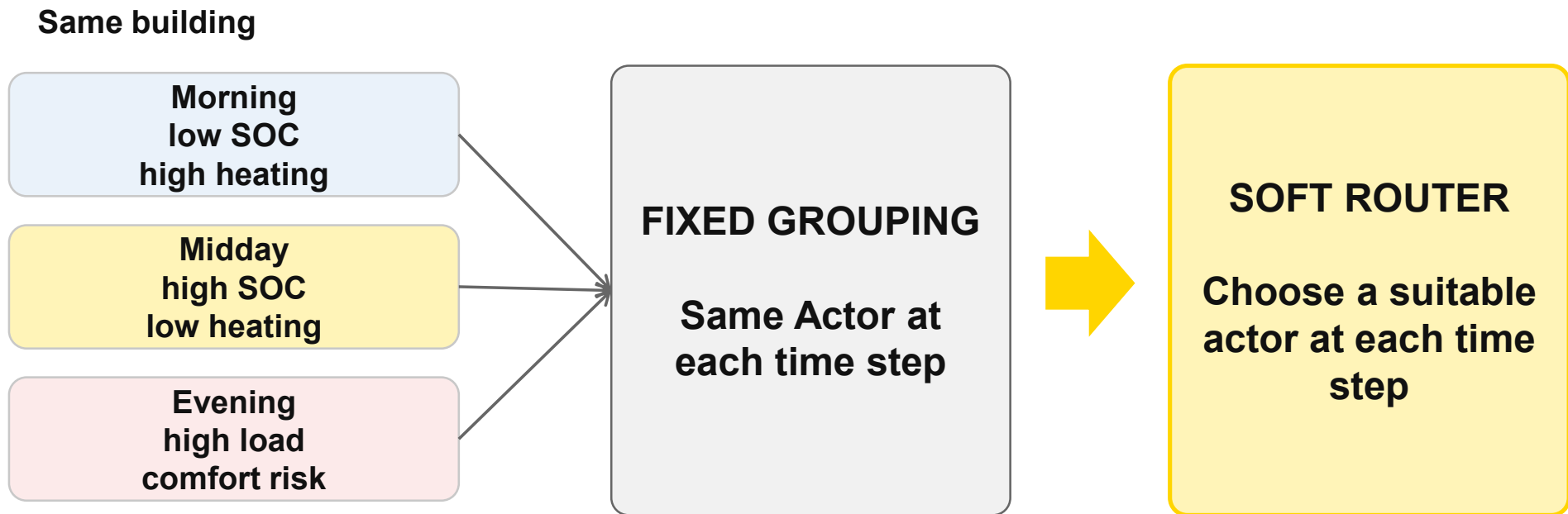
Compact features beat redundant inputs

3fA is selected for low bias, physical meaning, and stability



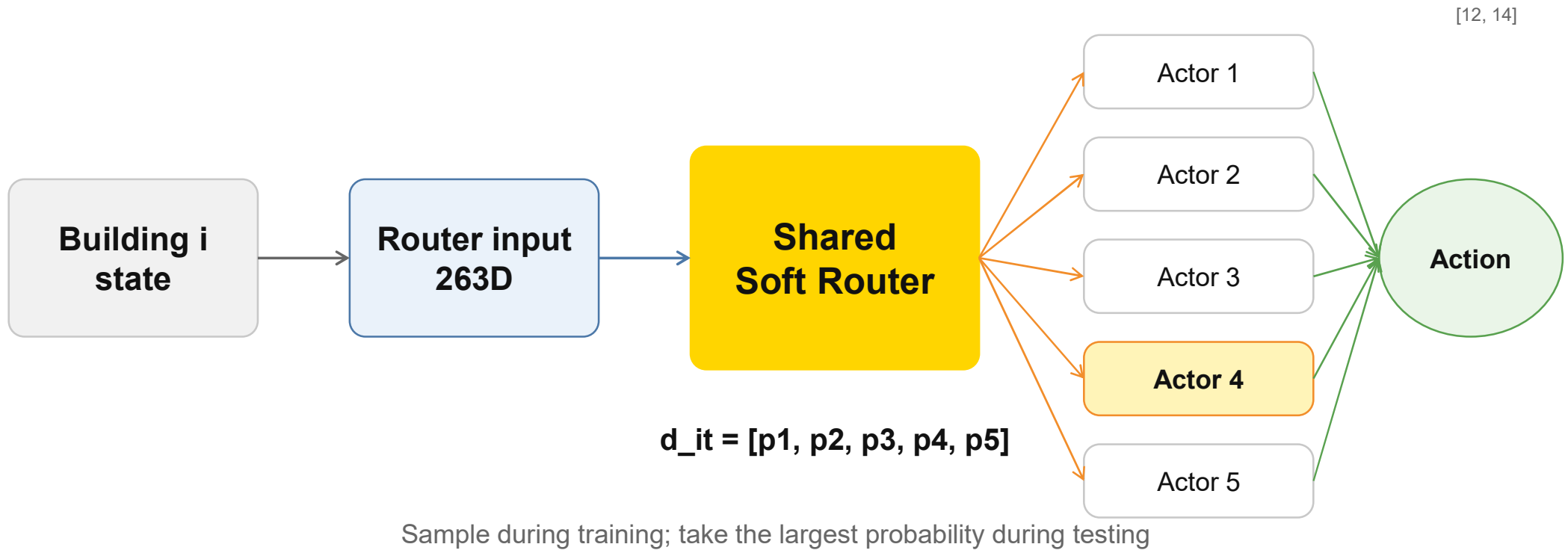
Strong actors still use fixed assignments

The same building still uses one actor in every state



Soft Router selects actors from current state

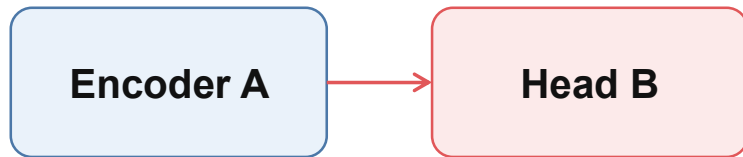
One actor distribution for each building and time step



The router must keep each actor complete

Encoder, communication path, and head belong together

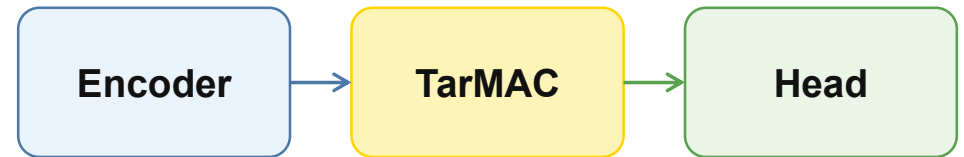
Head-only routing



Feature mismatch

Head B receives features that it did not learn with

Full-expert routing

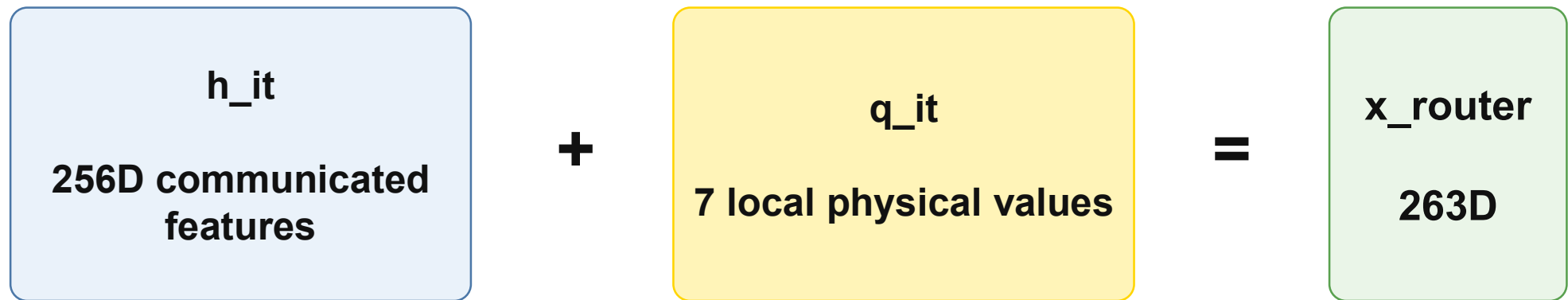


One complete expert branch

Matched full-expert run: 45.45% CV-RMSE

The router combines context and physics

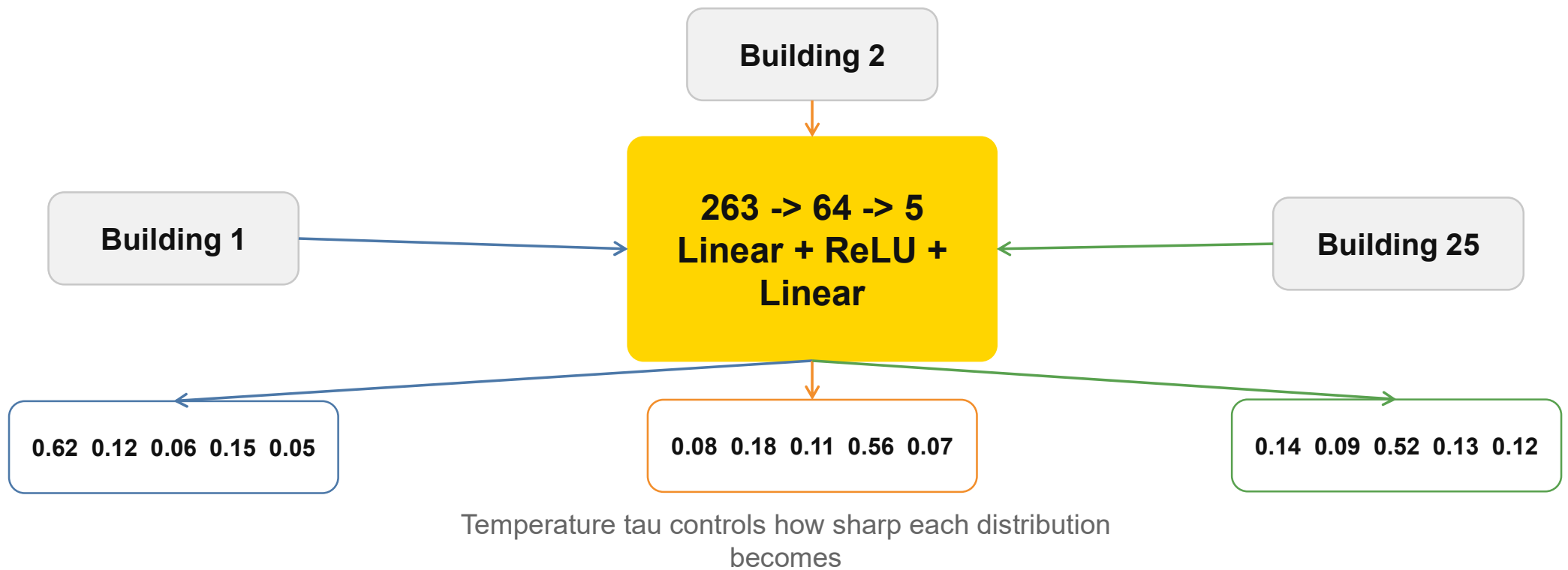
256 communicated features + 7 local values = 263 inputs



capacity | HVAC rating | SOC | charge space | discharge energy | NSL | heating demand

One shared router makes 25 local choices

Same network, different states, different probabilities



Soft routing forms a mixture policy

Changing router weights changes the full policy

[12, 14]

$$\pi_{\theta, \phi}(a_{it} \mid s_t) = \sum_{k=1}^K g_{itk}(\theta) \pi_{\phi_k}(a_{it} \mid s_t).$$

$$\pi(a_{i,t} \mid s_t) = \sum_k g_{i,t,k} * \pi_k(a_{i,t} \mid s_t)$$

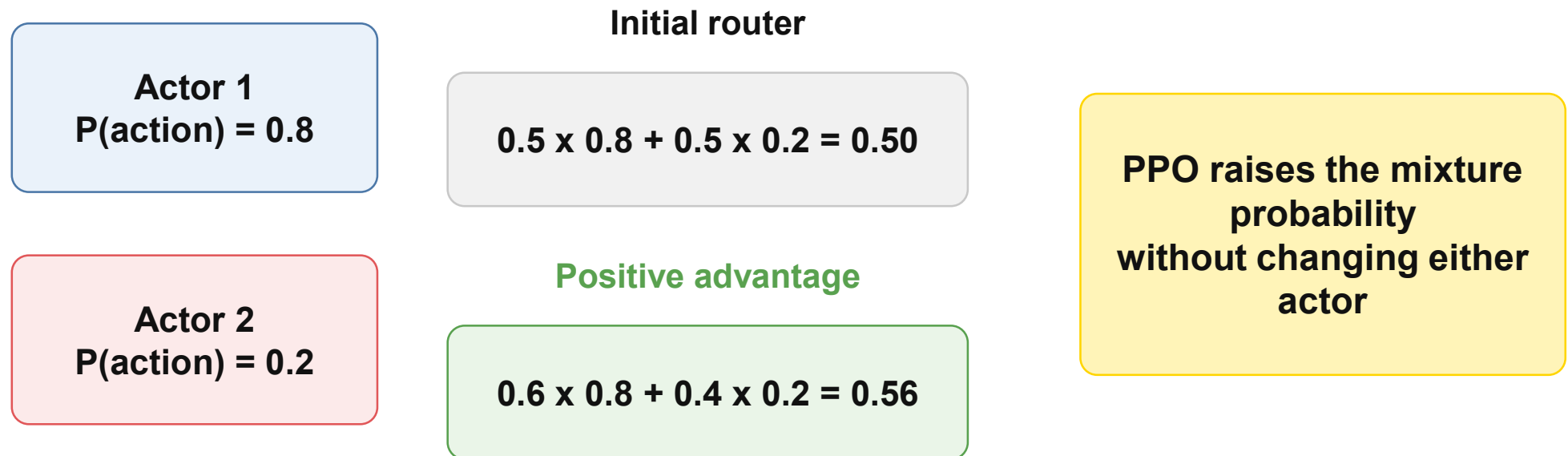
Router weights
 $g_{i,t,k}(\theta)$

Expert policies
 $\pi_k(a \mid s)$

Complete routed
policy
changes with θ

Frozen actors can still train the router

A simple probability example

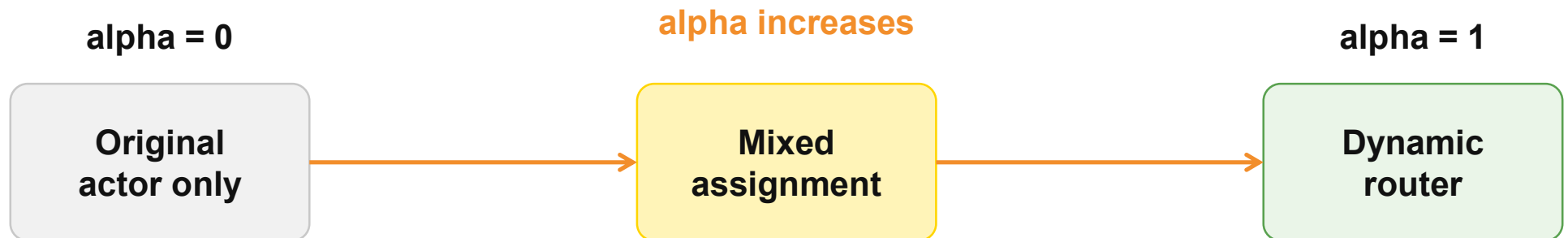


The router gains freedom step by step

The old group assignment is used as a safe starting point

$$g_{it} = (1 - \alpha)p_i + \alpha d_{it}.$$

$$g_{it} = (1 - \alpha) p_i + \alpha d_{it}$$



Lower tau: sharper choices | Higher tau: more exploration

Five controls reduce routing collapse

They keep more than one actor active

- | | | |
|---|--------------------------------|-----------------------------------|
| 1 | Probability sampling | Use probabilities during training |
| 2 | Old-group prior | Keep early actor diversity |
| 3 | Slow alpha increase | Avoid a sudden policy change |
| 4 | Entropy + balance | Reduce over-focused traffic |
| 5 | PPO clipping + freezing | Limit jumps and stop late drift |

No total one-expert collapse, but one branch can still be almost unused

Freezing the router stops late concentration

Matched runs share the same first 500 router-only episodes

Recorded point	Entropy	Mean largest gate	Largest share
Episode 1	1.223	0.602	30.6%
Episode 500	0.275	0.891	36.4%
Joint, episode 1000	0.171	0.934	51.7%
Frozen router, episode 1000	0.276	0.891	37.0%

Joint update

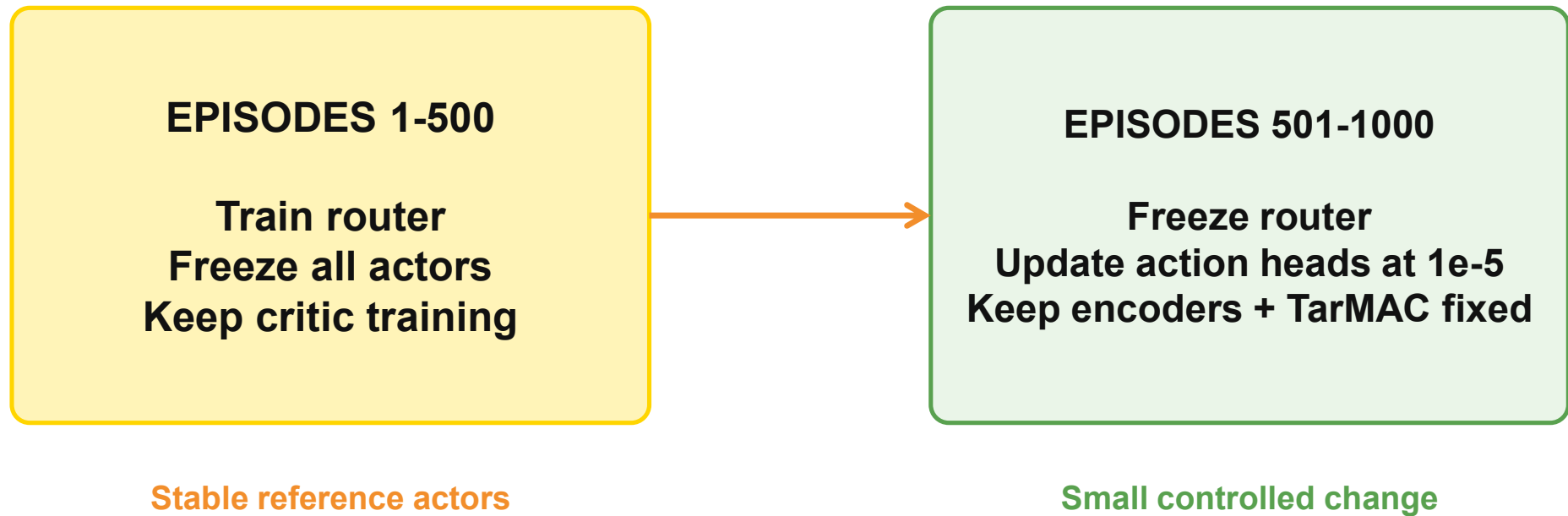
51.7%

Frozen router

37.0%

Final training uses two clear stages

Only one side changes at a time

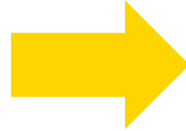


Stage separation removes two moving targets

Router learning and actor adaptation no longer chase each other

If both change together

**Router changes traffic
Actors change actions
PPO target keeps moving**



If stages are separated

**Router learns from fixed experts
Heads adapt under a fixed
router
Each stage has one clear target**

This is the main reason the stable-head version is easier to train

Full actors beat shared features

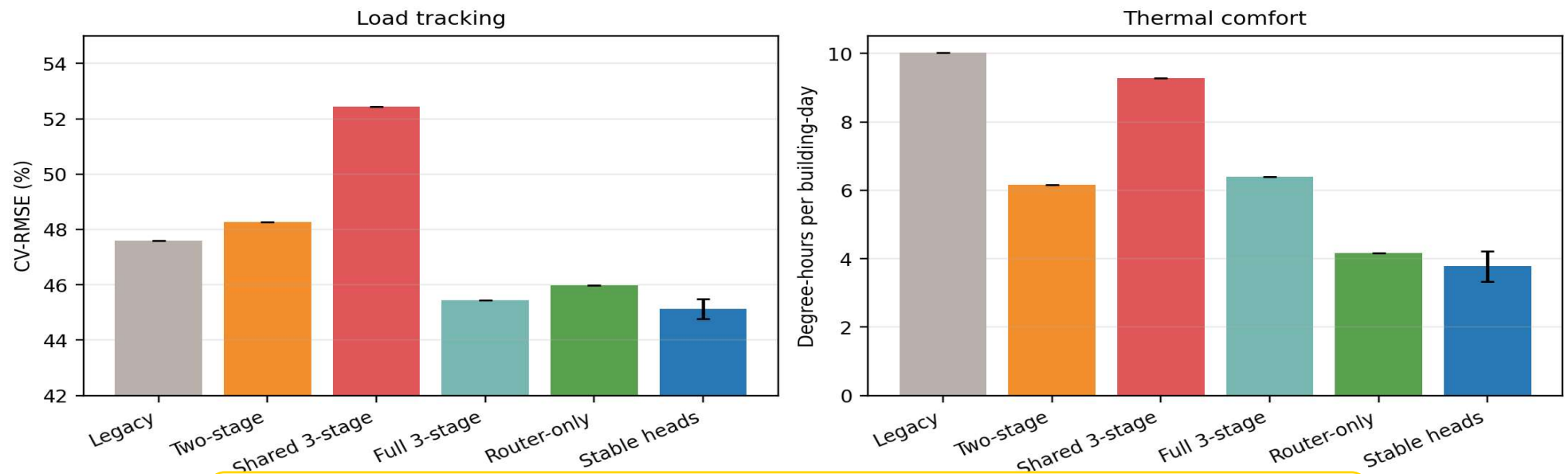
The router needs matched internal representations

Variant	CV-RMSE	NMBE	DH/day	Comfort
Shared encoder	52.44	7.48	9.27	31.35
Full actor	45.45	1.00	6.39	21.73
Stable heads	45.12 +/- 0.35	1.35 +/- 0.33	3.78 +/- 0.44	17.30 +/- 1.33

Keeping each actor complete is more important than sharing one feature space

Stable routing gives the best overall result

Ablation trend across the main development stages



Stable heads, 3 seeds
45.12 +/- 0.35% CV-RMSE | 3.78 +/- 0.44 DH/day

Three limits define the next work

The router is useful, but the research is not finished

1

Actor dependence

Train from weaker or random actors without collapse

2

Validation range

Use paired actor seeds, more climates, and larger portfolios

3

Building detail

Study actor traffic and long comfort violations

Goal: move from a stable enhancement toward independent adaptive routing

Soft Router improves three mean metrics

Better load shape and comfort, with a small bias trade-off

Controller	CV-RMSE	NMBE	DH/day	Comfort
Fixed 3fA, 5 runs	49.86 +/- 3.65	0.81 +/- 0.68	4.64 +/- 1.27	19.21 +/- 2.85
Stable heads, 3 seeds	45.12 +/- 0.35	1.35 +/- 0.33	3.78 +/- 0.44	17.30 +/- 1.33

-4.74 pts

mean CV-RMSE

-0.86

degree-hours/day

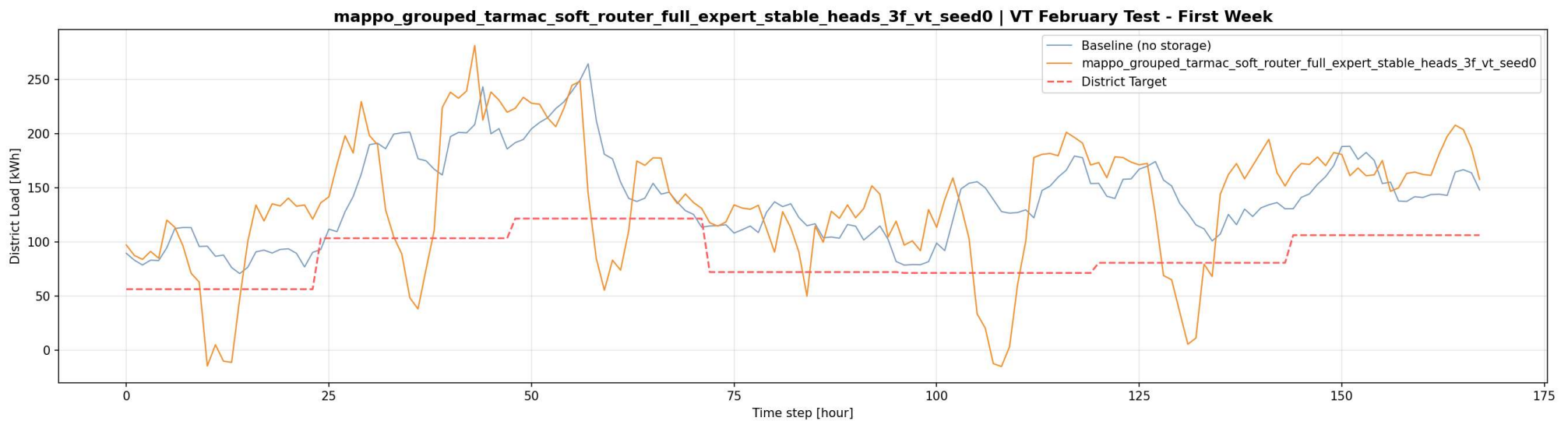
-1.91 pts

comfort exceedance

Main claim: state-based full-expert routing is a useful and stable enhancement

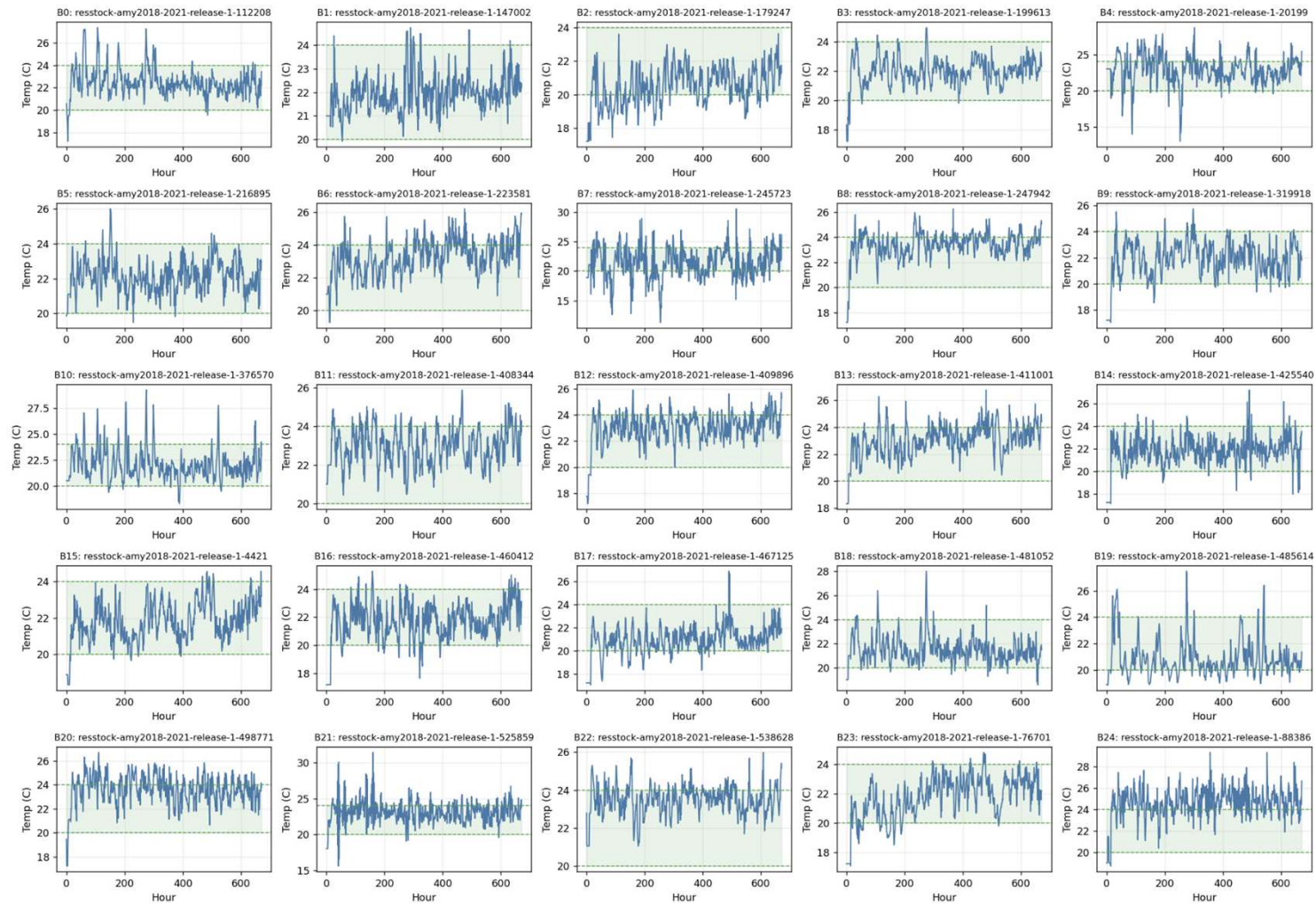
The controlled load moves closer to the target

Final stable router, first test week

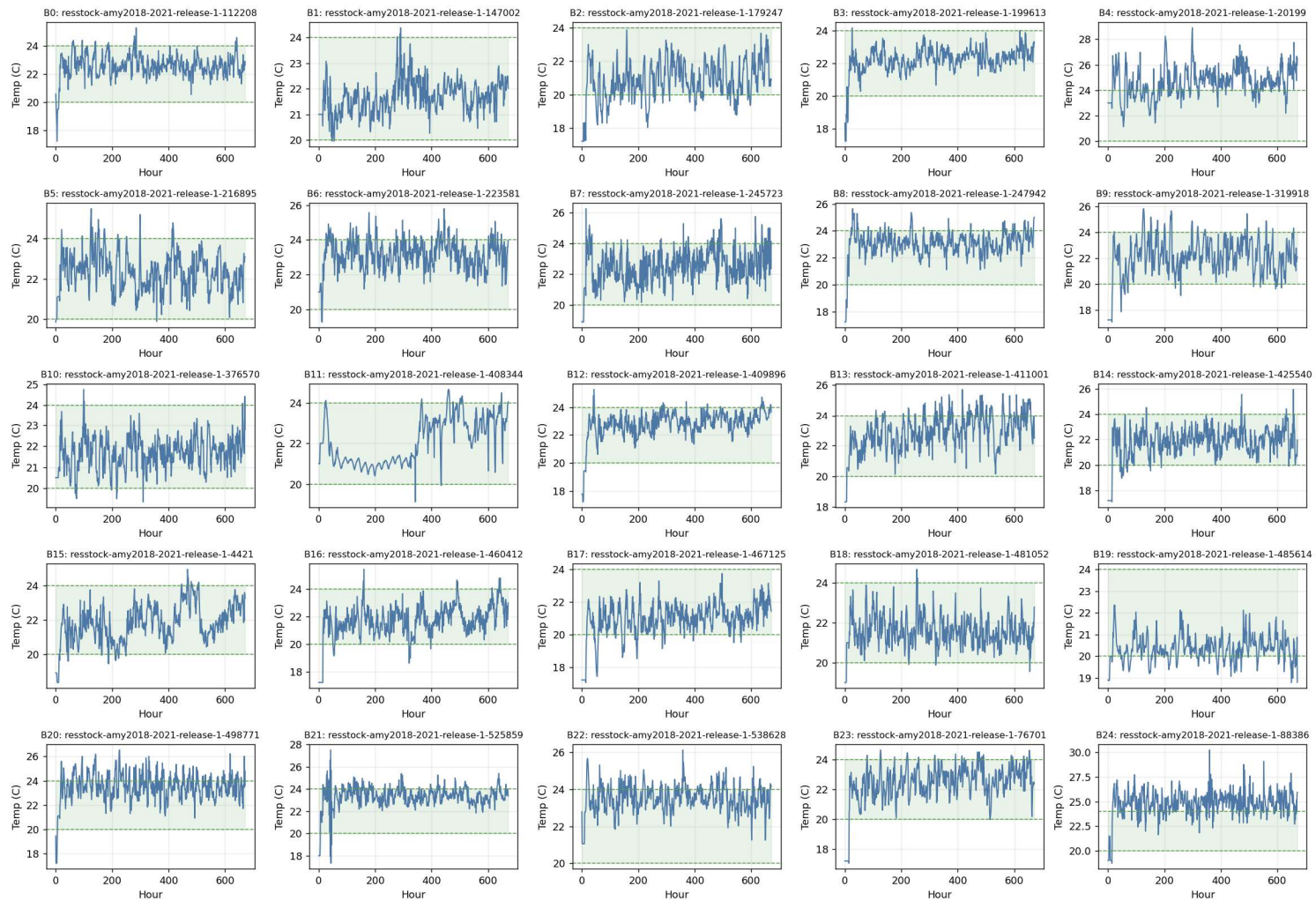


The shape improves, but some peaks and timing errors remain

Grouped MAPO (iarMAC) temperatures | VI | February | Full Test Window



Grouped MAPPO (IarMAC) temperatures | VT | February | full Test Window



Conclusion

Overall, this work improves grouped MAPPO through better communication, grouping, and feature selection. The main contribution is the Soft Router, which allows each building to choose a suitable complete actor at every time step. The final design achieves stable routing and a good balance between load tracking and comfort.

References

Sources cited in this presentation

[5] Das et al. TarMAC: Targeted Multi-Agent Communication. ICML, 2019.

[11] Nweye et al. CityLearn v2: Energy-flexible, resilient, occupant-centric, and carbon-aware management of grid-interactive communities. JBPS, 2024.

[12] Obando Ceron et al. Mixtures of Experts Unlock Parameter Scaling for Deep Reinforcement Learning. ICML, 2024.

[13] Peng, Long, and Ding. Feature Selection Based on Mutual Information: Max-Relevance and Min-Redundancy. IEEE TPAMI, 2005.

[14] Puigcerver et al. From Sparse to Soft Mixtures of Experts. ICLR, 2024.

[16] Vazquez-Canteli et al. CityLearn: Standardizing Research in Multi-Agent Reinforcement Learning for Demand Response and Urban Energy Management. 2020.

[18] Ward. Hierarchical Grouping to Optimize an Objective Function. JASA, 1963.

All experiment values and figures are from the user-provided Progress Report 2.